Do Bayesian Neural Networks Actually Behave Like Bayesian Models?

Gábor Pituk, Vik Shirvaikar, Tom Rainforth

Reading Group Presentation

March 6, 2026

Background and Motivation

- **Bayesian Neural Networks (BNNs):** BNNs extend Bayesian inference to neural networks by placing prior distributions on model parameters and updating them based on observed data.
- **Theoretical Promises:** They promise benefits such as principled uncertainty quantification, robustness to distribution shift, and coherent belief propagation.
- **The Intractability Problem:** Exact Bayesian inference in deep learning is fundamentally intractable, necessitating the use of approximate inference methods (e.g., VI, Laplace, SWAG, MCMC).
- **Core Motivation:** Why does it matter if these approximate BNNs do not behave like true Bayesian models?
  - If BNNs fail to respect Bayesian principles, we cannot trust that better priors or improved inference will translate to actual performance gains.
  - Downstream applications like active learning (e.g., BALD) and continual learning fundamentally assume the model will be updated in a true Bayesian manner without forgetting old data.

Core Research Objectives and Key Findings

- **Research Objective:** To systematically investigate the adherence of practical BNN algorithms to the expected behavior of true Bayesian models, moving beyond traditional static predictive performance metrics.
- **Overall Conclusion:** Practical BNNs fail to preserve key features of Bayesian inference across synthetic regression and real-world classification tasks.
- **Three Key Findings:**
  1. **Lack of Functional Variability:** Approximate posteriors lack the expected functional variability and complexity of true Bayesian posteriors in regions without data.
  2. **Sequential Incoherence:** Sequential inference using BNNs fails to propagate information coherently, leading to catastrophic forgetting of older data.
  3. **Predictive Incoherence:** BNN predictive updates are not well-calibrated and fail to retain a self-consistent amount of uncertainty, violating fundamental information-theoretic properties.

Bayesian Neural Networks

- In standard neural networks, parameters θ are fixed; learning is a point estimate.
- In BNNs, weights and biases are random variables with a prior $p(\theta)$.
- Posterior via Bayes rule: $p(\theta \mid D) \propto p(D \mid \theta)\, p(\theta)$.
- Predictions are made by marginalising over the posterior:

$$p(y \mid x, D) = \int p(y \mid x, \theta)\, p(\theta \mid D)\, d\theta.$$

- Exact inference is typically intractable $\Rightarrow$ approximate inference: variational inference, MCMC, SWAG, etc.

How Are BNN Parameters Optimized?

- **Goal**: approximate the posterior
$$p(\theta \mid D) \propto p(D \mid \theta)p(\theta)$$
instead of just minimizing loss at a point.

- **Main approaches**:
  - **Variational Inference (VI)**: optimize $q_\phi(\theta)$ via ELBO
$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi}[\log p(D \mid \theta)] - \mathrm{KL}(q_\phi \| \, p(\theta))$$
with reparameterization $\theta = \mu + \sigma \odot \epsilon$.
  - **MCMC** (e.g. HMC, SGLD): asymptotically exact, but expensive.
  - **Laplace Approximation**: Gaussian around MAP estimate.
  - **SWAG**: approximate posterior via SGD trajectory.

- **Key insight**: Training learns a *distribution over parameters*, not just a single point estimate.

Variational Inference and ELBO

- Approximate the posterior with a tractable family $q_\phi(\theta)$ by minimising $\mathrm{KL}\big(q_\phi \parallel p(\theta \mid D)\big)$.
- Evidence lower bound (ELBO):

$$\mathcal{L}(\phi) = \mathbb{E}_{q_\phi}[\log p(D \mid \theta)] - \mathrm{KL}\big(q_\phi(\theta) \parallel p(\theta)\big).$$

- The first term encourages fitting the data, the second regularises toward the prior.
- Mean–field VI often underestimates uncertainty but is scalable.

Reparameterization Trick

- **Problem:** In VI we need gradients of

$$\mathbb{E}_{q_\phi(\theta)}[\log p(D \mid \theta)],$$

  but sampling $\theta \sim q_\phi(\theta)$ breaks the computational graph. Backpropagation cannot flow through a random draw.
- **Solution:** Rewrite sampling as a deterministic transform of fixed noise:

$$\theta = \mu_\phi + \sigma_\phi \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

  - ϵ: pure randomness, fixed distribution.
  - μ_ϕ, σ_ϕ: trainable, gradients flow.

Laplace Approximation

- **Concept**: Provides a local Gaussian approximation of a distribution around its Maximum A Posteriori (MAP) estimate.
- **Mathematical Derivation**: A second-order Taylor expansion of the log-density around θ_{MAP} (where the gradient vanishes):

$$\log f(\theta) \approx \log f(\theta_{MAP}) + \frac{1}{2}(\theta - \theta_{MAP})^\top \nabla^2 \log f(\theta_{MAP})(\theta - \theta_{MAP})$$

  Normalizing this yields a Gaussian posterior:

$$f(\theta) \approx \mathcal{N}\left(\theta; \theta_{MAP}, [-\nabla^2 \log f(\theta_{MAP})]^{-1}\right)$$

- **Practical BNN Scaling**:
  - The exact Hessian is quadratically large and prohibitive for deep networks.
  - Alternatives: Use the Fisher Information Matrix $I(\theta)$ or Empirical Fisher $\hat{I}(\theta)$.
  - To fit in memory, practitioners use restrictive Diagonal approximations or block-diagonal approaches like K-FAC.

Stochastic Gradient Langevin Dynamics (SGLD)

- **Concept**: A practical Markov Chain Monte Carlo (MCMC) sampling scheme based on continuous-time Langevin diffusion.
- **Mathematical Formulation**: The continuous dynamics $d\theta_t = -\nabla \log \pi(\theta|\mathcal{D})dt + \sqrt{2}B_t$ is simulated using an Euler discretization scheme:

$$\theta_{t+1} = \theta_t - \epsilon_t \nabla \log \pi(\theta_t|\mathcal{D}) + \sqrt{2\epsilon_t}\xi_t$$

  where ϵ_t is the learning schedule and $\xi_t \overset{iid}{\sim} \mathcal{N}(0, I)$.
- **Stochastic Batches**: For computational efficiency in BNNs, the exact gradient $\nabla \log \pi(\theta_t|\mathcal{D})$ is replaced with a stochastic estimate over mini-batches of size B.
- **Trade-offs**: To maintain computational efficiency, SGLD traditionally skips the Metropolis-Hastings accept-reject step. The resulting bias is theoretically compensated for by decreased variance.

Related Work: Critiques and Evolving Perspectives

- **Fundamental Critiques**: BNNs are often criticized for using uninformative priors that fail to reflect true beliefs. Furthermore, the fidelity of approximate inference is routinely questioned due to theoretical pathologies. Consequently, BNNs often underperform under covariate shift and model misspecification.
- **The "Cold Posterior" Effect**: Artificially sharpening the posterior (using a temperature $T < 1$) often yields better predictive performance than the true Bayes posterior ($T = 1$). This suggests underlying issues with the model setup, data augmentation, or the ability of samplers to explore distinct modes.
- **Task-Specific Failures**: In practical downstream applications, approximate BNNs fall short in active learning and exhibit catastrophic forgetting in continual learning contexts.
- **A Paradigm Shift**: Researchers increasingly argue for a pragmatic view, treating BNNs as practical probabilistic algorithms with good static predictive performance rather than strictly true Bayesian models.
- **This Paper's Niche**: While prior studies heavily focused on static predictive performance and convergence diagnostics , this paper shifts the focus to systematically testing how BNNs behave regarding the theoretical properties of Bayesian models.

Exp 1: Function Space Variability (Methodology)

Experimental Purpose:

- Do practical BNN posteriors capture true functional variability and epistemic uncertainty in regions without data?
- Do they actually "know what they don't know"?

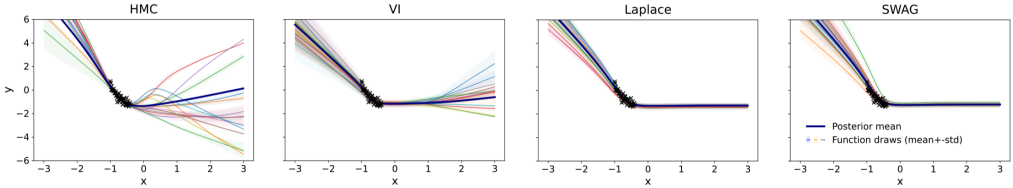
Experimental Workflow:

1. **Data Generation**: Create a 1D synthetic regression task with a "data gap" in the middle and no data at the edges.
2. **Model Training**: Train HMC, VI, Laplace, and SWAG on a heteroscedastic likelihood to output both mean and variance.
3. **Uncertainty Quantification**: Use the BALD (Bayesian Active Learning by Disagreement) score to evaluate epistemic uncertainty. BALD measures the expected information gain—essentially, how much a new observation at a given location would teach the model about its underlying parameters.
4. **Evaluation**: BALD acts as a "lie detector" for model confidence. High scores mean the model correctly recognizes its own ignorance in unseen regions. Near-zero scores in areas without data expose dangerous, blind overconfidence.

Exp 1: Posterior Function Space Variability (Figure 1)

What the Figure Shows:

- **Setup**: A 1D synthetic regression task with a "data gap" in the middle and no data at the edges.
- **Visual Elements**: The black crosses represent observed data points. The solid black line is the posterior mean (BMA).
- **Comparison**: The colored lines are individual function draws sampled from the approximate posteriors of HMC, VI, Laplace, and SWAG.

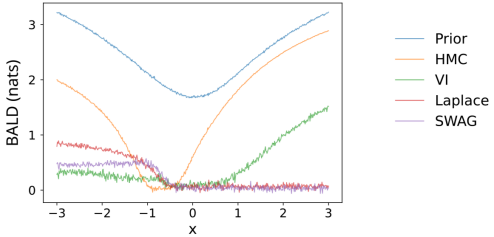


Takeaway Message: While the posterior means look similar, the underlying distributions are drastically different. HMC captures complex functional diversity in regions without data ($x > 0$). VI lacks qualitative diversity, and Laplace/SWAG completely collapse into simple parallel shifts of the MAP estimate, failing to capture meaningful epistemic uncertainty.

Exp 1: Information Gain & Uncertainty (Figure 2)

What the Figure Shows:

- **Metric**: Plots the BALD (Bayesian Active Learning by Disagreement) scores across the input space x for the models from Figure 1.
- **Axes**: The Y-axis measures the expected information gain (in nats) regarding model weights if we were to observe a new data point at location x.



Takeaway Message: HMC correctly expects the highest information gain far away from existing observations (where uncertainty should be highest). In contrast, Laplace and SWAG are blindly confident: they expect near-zero information gain in completely unseen regions ($x > 0$), making them highly dangerous for active learning.

Exp 2: Sequential Belief Updates (Methodology)

Experimental Purpose:

- Are sequential belief updates coherent in BNNs?
- If we feed data to the model in chunks, does it perfectly retain old information, or does intermediate training cause catastrophic forgetting?

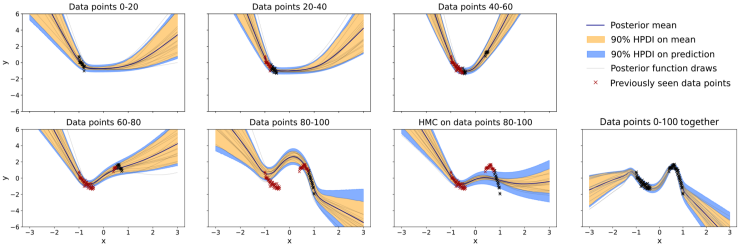
Experimental Workflow & Math:

1. **Theorem Setup**: Pure Bayesian coherence dictates invariance to observation order: $\pi((\theta|D^{(1)})|D^{(2)}) = \pi(\theta|D^{(1)} \cup D^{(2)})$.
2. **Data Splitting**: Split datasets (e.g., CIFAR-10) into two chunks: $D^{(1)}$ and $D^{(2)}$.
3. **Intermediate Training**: Train an intermediate posterior on the first split: $\pi_{D^{(1)}}(\theta)$.
4. **Sequential Inference**: Pass the intermediate posterior as the prior to train on the second split, yielding the sequential posterior: $(\pi_{D^{(1)}})_{D^{(2)}}(\theta)$.
5. **Evaluation**: Compare the BMA predictive performance of the sequential posterior against a single-step posterior trained on all data at once.

Exp 2: Sequential Updates in Regression (Figure 3)

What the Figure Shows:

- **Setup**: A BNN (using VI) is trained sequentially on 5 distinct chunks of the synthetic regression data.
- **Visual Elements**: Black dots are the data points currently being trained on; red crosses are data points from previous steps. The shaded regions represent the 90% High Probability Density Intervals (HPDI) for the mean and observations.

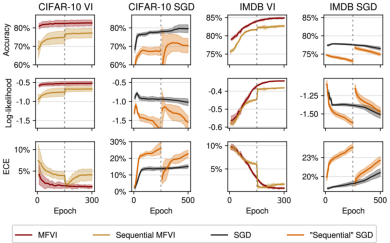


Takeaway Message: The VI algorithm exhibits catastrophic forgetting. As it learns new data chunks, it fundamentally alters its predictive shape, failing to retain the functional knowledge required to fit the older, previously seen data points.

Exp 2: Sequential Updates in Classification (Figure 4)

What the Figure Shows:

- **Setup**: BMA performance metrics on CIFAR-10 and IMDB.
- **Comparison**: Compares "MFVI" (trained on all data in a single step) against "Sequential MFVI" (trained sequentially on splits).
- **Metrics**: Tracks Accuracy, Log-likelihood, and Expected Calibration Error (ECE) over training epochs.



Takeaway Message: Under true Bayesian theory, the sequential posterior should equal the single-step posterior. However, Sequential MFVI significantly and consistently underperforms Single-step MFVI across all metrics. The use of VI for sequential inference is not justified by Bayesian motivations alone.

Exp 3: Theoretical Core – Bayesian Updates as Martingales

1. **Intuition: a "fair game"**
   - A rational agent cannot systematically predict directional belief shifts.
   - If such a shift were predictable, it should be incorporated *now*.
2. **Math: law of iterated expectations**
   - Today's posterior mean: $\bar{\theta}_{n-1} = \mathbb{E}[\Theta \mid Y_{1:n-1}]$.
   - Expected tomorrow mean:
   $$\mathbb{E}[\bar{\theta}_n \mid Y_{1:n-1}] = \mathbb{E}[\mathbb{E}[\Theta \mid Y_{1:n}] \mid Y_{1:n-1}]\,.$$
   - Tower property gives
   $$\mathbb{E}[\mathbb{E}[\Theta \mid Y_{1:n-1}, Y_n] \mid Y_{1:n-1}] = \mathbb{E}[\Theta \mid Y_{1:n-1}] = \bar{\theta}_{n-1}.$$
3. **Experimental implication**
   - **Doob (1949)**: exact Bayesian updates form a martingale.
   - Therefore, averaging posteriors retrained on self-imputed pseudo-data should recover the initial posterior.
   - If the average prediction drifts, the update rule is not truly Bayesian.

Exp 3: Predictive Coherence (Methodology)

Experimental Purpose:

- Are predictive updates well-calibrated?
- If a model generates pseudo-data based on its own uncertainty and retrains on it, does its average prediction remain stable (Martingale property), or does it hallucinate new information?

Experimental Workflow & Math:

1. **Initial Posterior**: Train on labeled data (x, y) to get $\pi_{(x,y)}(\theta)$.
2. **Predictive Resampling**: Impute pseudo-labels $\hat{y}^*$ for unlabeled data x^* via the marginal predictive: $Y^* \sim \int p(y^*|x^*, \theta)\pi_{(x,y)}(d\theta)$.
3. **Imputed Posteriors**: Retrain on mixed data to get $\pi_{(x,y)\cup(x^*,\hat{y}^*)}(\theta)$.
4. **Empirical Martingale Posterior**: Average N retrainings:

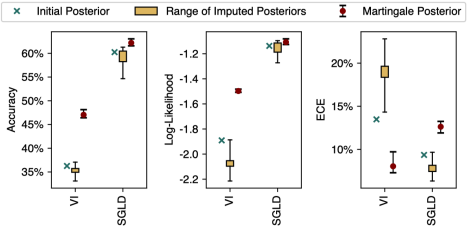
$$\hat{MP}^{(N)}_{(x^*,Y^*)}\pi_{(x,y)}(\theta) = \frac{1}{N}\sum_{j=1}^{N}\pi_{(x,y)\cup(x^*,\hat{y}_j^*)}(\theta)$$

5. **Theorem Test**: A coherent Bayesian model requires the Martingale expectation to match the initial belief: $\mathbb{E}[\pi(\theta|Y^*)] = \pi(\theta)$.

Exp 3: Martingale Posterior Performance (Figure 7)

What the Figure Shows:

- **Setup**: Evaluates predictive coherence on CIFAR-10.
- **Legends**:
  - **Green Cross**: The initial posterior trained on real data.
  - **Yellow Boxes**: The range of performance for individual models retrained on pseudo-labels (Imputed Posteriors).
  - **Red Dot**: The "Martingale Posterior" (the average ensemble of all yellow box models) with 95% bootstrap CI.

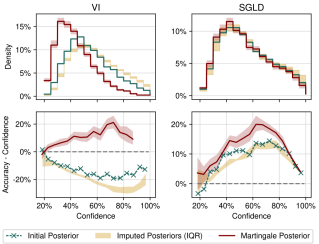


Takeaway Message: Imputing self-generated labels should not introduce new information; the red dot *should* overlap with the green cross. Because it drastically deviates, Bayesian coherency is falsified. Ironically, the Martingale ensemble outperforms the initial BMA due to standard ML ensembling effects.

Exp 3: Predictive Uncertainty Calibration (Figure 8)

What the Figure Shows:

- **Top Row (Density)**: Distribution of the models' confidence (the maximum predicted class probability).
- **Bottom Row (Calibration Curves)**: Plots (Accuracy minus Confidence) against Confidence. A perfectly honest model hugs the dashed zero line; dipping below means the model is overconfident.



Takeaway Message: The initial VI posterior (green line) is severely overconfident. Retraining on pseudo-labels makes individual models even more overconfident (yellow area). However, averaging them via the Martingale Posterior (red line) smooths out these extreme biases, resulting in a slightly under-confident but significantly more accurate and better-calibrated predictor.